wcurl - one year later



DebConf 25

Brest, France

Slides available at samueloph.dev/slides

About me

- Samuel Henrique <samueloph>
- Debian Developer since 2018
  - Contributing on personal time, as a volunteer (not during work hours)
- Member of Security Tools Packaging Team
- Maintains curl, rsync, shellcheck...
- Mentorship for Debian packaging
- Senior System Development Engineer in Amazon Linux, security team
- Debian/Linux, Python, Rust, Bash, Security
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Inception

- Idea on the way to curl-up 2024
- Discussed it for the first time in the second day of curl-up 2024, over lunch, called it “curld”
  - “*worth a try*”
- Discussed it with Debian curl maintainters and other Brazilian Debian Developers
  - “*curld sounds like a daemon*”, Charles Melara suggested “wcurl” instead
- Sergio Durigan Junior <sergiodj> ported it from bash to POSIX-compliant shell script
  - Besides working on a bunch of other nice features
- First release in 2024-07-02

Public reception

- Majoritarily good, *but...*
- *“A simplest alias `dl='curl -O'` just does the same for me.”*
- *“wcurl seems neat, but the idea that curl is so complex we can only fix it by adding *more* code in the form of a shell script ... bothers me. Seems like the lessons of Unix keep being forgotten.”*
- *“This happens so often. A developer could just read a man page or learn something deeply by example, but they don't. Instead they struggle for the rest of their lives in ignorance, even going so far as to adopt a hack to avoid having to learn a new thing. They will go far out their way to not learn a new thing, and over time, cause themselves untold misery.”*

Contributions

- Non GNU/Linux support - “tssva”
  - https://news.ycombinator.com/item?id=40873627
- HTTPS as default protocol - “heinrich5991”
  - https://github.com/curl/wcurl/issues/1#issuecomment-2208468484
- Disable curl’s globbing parser - “gnomon-”
  - https://github.com/curl/wcurl/pull/9
- Default filename index.html - Ryan Carsten Schmidt
  - https://salsa.debian.org/debian/wcurl/-/merge_requests/4
- Percent-decode resulting filename - Ryan Carsten Schmidt
  - https://salsa.debian.org/debian/wcurl/-/merge_requests/4
- Logo - Daniel Stenberg
- Allow “-o” and “-O” to be used without whitespaces (e.g.: “-oNAME”) - “MithicSpirit”, who uses Arch BTW
  - https://github.com/curl/wcurl/issues/42

Availability

- The major convenience factor comes from being installed automatically with curl
  - In **Debian 13** and **12-backports** since June 2024 (wcurl creation)
  - In **Ubuntu 24.10** since August 2024
  - In **Amazon Linux 2** and **Amazon Linux 2023** since April 2025
  - In **Alpine Linux 3.22** since May 2025
  - In **Arch Linux** since May 2025
  - In **Fedora 43** since May 2025
  - In **Gentoo** since May 2025
  - In **Homebrew** since May 2025
  - In **MSYS2** since May 2025
  - In **Slackware** since May 2025
  - In **FreeBSD** since June 2025
- Shipped as separate package “wcurl”, not automatically installed with curl
  - **OpenSUSE**
  - **NixOS**
- Installed as part of curl upstream since May 2025 (8.14.0)

Looking back

- wcurl is an extremely simple solution that does one use case well
- curl developers' support was essential for the adoption
- The right place and in the right time
- Me and other contributors spent hours discussing small design decisions, such as what should the "--curl-options" parameter be called
  - It really paid off in the end
- Other people's contributions made a difference

What’s Next

- *“Have you heard about wcurl?”*
  - There are more wcurl stickers than wcurl lines of code
- The current version of wcurl has 0 maintenance cost
  - One more release for fixing typos (#51)
- About a compiled wcurl
  - No concrete plans
  - Fancier features, like wget
  - More portable but also less portable (requires compilation)
  - If it happens, it’s likely to co-exist with the script wcurl

Muito obrigado!



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